LAB 02 PSEUDOCODE ALGORITHMS AND FLOWCHARTS

Algorithms

The word Algorithm means "a process or set of rules to be followed in problem-solving operations". Therefore, Algorithm refers to a set of rules or instructions that step-by-step define how a work is to be executed in-order to get the expected results.

Pseudocode

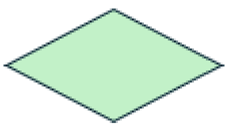
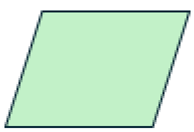
Pseudocode is a way of expressing an algorithm using syntax that is independent of any particular programming language. The algorithm can then be coded in a suitable programming language.

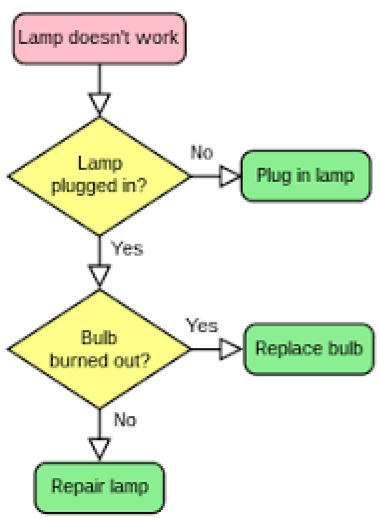
Pseudocode for making a cup of tea:

1. START
2. Fetch a tea cup
3. Boil some water
4. Place a tea bag into the cup
5. Pour on boiling water
6. Stir tea bag
7. Remove tea bag
8. END

Flow Chart

Flowchart is a graphical representation of an algorithm. Programmers often use it as a program-planning tool to solve a problem. It makes use of symbols which are connected among them to indicate the flow of information and processing.

Symbol	Name	Function
	Oval	Represents the start or end of a process
	Rectangle	Denotes a process or operation step
	Arrow	Indicates the flow between steps
	Diamond	Signifies a point requiring a yes/no
	Parallelogram	Used for input or output operations



Tasks

1. Write an **algorithm** for determining the type of a treaingle based on the length of its 3 sides.
    - Equilateral Triangle: All three sides are of equal length.
    - Isosceles Triangle: Two sides are of equal length, while the third side is different.
    - Scalene Triangle: All three sides are of different lengths.
2. Write **pseudocode** that takes 3 numbers as input and sorts them in descending order.
3. Write **algorithm** and create **flowchart** for an ATM system with the following options
    - Check Balance
    - Withdraw Cash
    - Change PIN
4. Write **pseudocode** that takes the input values for a, b and c and solves the equation given below to find the values of x.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

5. Write a **pseudocode** and make a **flowchart** for a program that searches for a specific file across all drives and folders in the available file system. The program will take the filename as input and search every directory recursively until it finds the file or confirms that it does not exist.